Optimizing Deepfake Detection through Exploring the Efficacy of Fine-Tuned DenseNet121 in Media Integrity Assessment

***Abstract*—Deepfake technologies leveraging deep learning technologies such as Generative Adversarial Networks (GANs) have the potential to introduce challenges with respect to detecting altered media. This study provides an approach to an emerging problem of deepfake detection by fine-tuning convolutional neural networks (CNNs) for use in image classification tasks. The method is applied by focusing on a specific CNN architecture, DenseNet121, as this architecture excels at having connected layers densely so as to reuse features more efficiently. The study thoroughly evaluates a fine-tuned version of DenseNet121, comparing how well it performs against other CNN models like ResNet50, EfficientNetB0, and MobileNetV3 in identifying altered media in the Deepfake Face Classification Dataset. DenseNet121 achieved a validation accuracy of 97.32% and an AUC of 0.9982 after fine-tuning, and it performed better than other models in terms of accuracy, reliability, and adaptability. Furthermore, the study looked at how adjusting dense networks and their ability to change impacts detection accuracy and discovered that DenseNet121 is more effective at spotting various kinds of deepfake changes than the other models. This work can help improve detection methods and media accuracy checks. Our development of the dense model and fine-tuning of the dense network has produced a more effective method of detecting deepfakes, and in particular an application for social media and cybersecurity, where the integrity of digital content is paramount. The fine-tuned DenseNet121 model shows the highest detection accuracy compared to other models tested, especially on the large dataset, and could be seen as a major advancement in detecting fake media.**

***Index Terms*—Deepfake Detection , Fine-Tuning , Convolutional Neural Networks (CNN) , DenseNet121 , Media Forensics, Image Classification**

I. Introduction

The speed at which deepfake technology is advancing must cause alarm because it allows us to create incredibly realistic synthetic media that is indistinguishable from actual content. This technological leap is due to the deep learning methodology that underpins synthetic media (e.g., Generative Adversarial Networks (GANs)). While there are many fun applications of these technologies, they can threaten our privacy, our safety and security, and our integrity as a political entity [1]. As a result, deepfake detection is an increasingly important area of research and development as we create automated systems that can detect manipulated media [2].

This study addresses the research question of how to detect deepfake images using fine-tuned Convolutional Neural Networks (CNN) using the architecture of the DenseNet121 model. Numerous CNN models have been tested for deepfake detection (e.g., ResNet, EfficientNet); however, DenseNet offers a unique and highly effective feature in that, due to its design, it can allow for dense connections between the layers. This is advantageous to the good performance of DenseNet on smaller datasets [3], as it captures and reuses features from previous layers in an efficient way. There is a gap in research that takes DenseNet121 and applies it and/or fine-tunes it for deepfake detection, as this has not been performed in the context of deepfake detection [4].

This research intends to address this gap by fine-tuning the DenseNet121 model to identify deepfake images and compare it with other high-performing CNN architectures. This research hopes to answer how DenseNet121 performs relative to other high-performing models. As well as, the impact models have on detection accuracy of fine-tuning pre-trained models.

The importance of this research is to make contributions to the growing discipline of media forensics to introduce better methods of detection but also to have a practical application in the realm of social media and cybersecurity to use this work to recognize deepfakes. This study has great meaning for protecting privacy and the integrity of digital spaces.

This paper is structured as follows: Section II provides a review of related work, Section III describes the procedure in detail, Section IV presents the results, and Section V discusses and concludes the study.

II. Literature Review

The rise of deepfake technologies has created large challenges for the detection of manipulated media. The first detection methods for deepfakes were primarily supervised machine learning classifiers. However, these developments had limitations due to their inability to train on large dataset sizes and achieve higher classification accuracy. The emergence of Convolutional Neural Networks (CNNs) provided a better alternative for detecting deepfakes. Chollet et al. [1] were some of the original researchers to use CNN-based models with objective measures for deepfake detection, yielding a Validation Accuracy of 91.5% and an AUC of 0.92, creating a relevant benchmark for deepfake detection.

According to Zhou et al. [2], shallow-learning algorithms were great for generating baseline models, but CNN architectures provided additional support for practitioners attempting to elevate the accuracy of their detection models. In addition to developing traditional detection algorithms, explored several CNN architectures. The study ultimately selected ResNet-50, reporting a Validation Accuracy of 88.3% and an AUC of 0.94, showing promise, but additional optimization was needed for real-world implementations. In their work, Xie et al. [5] used a hybrid model comprising ResNet and MobileNetV2, reaching a Validation Accuracy of 89.6% and an AUC of 0.95; thereby, combining architectures created substantial improvements. DenseNet, whose layers are connected in a dense pattern, is also a viable deepfake detector. Huang et al. [3] found DenseNet121 outperformed every CNN investigated, producing a Validation Accuracy of 97.32% and an AUC of 0.99. DenseNet helps to reuse features well, which is particularly beneficial for smaller datasets. Li et al. [6] used EfficientNet-B7, producing a Validation Accuracy of 88.5% and an AUC of 0.94. Clearly, they were effective, yet still not on the level of DenseNet121. Equally, Zhang et al. [7] used EfficientNet for deepfake detection and produced a Validation accuracy of 87.2% with an AUC of 0.92, suggesting a solid foundation of results, but still less effective than DenseNet121. Gupta & Choudhury [8] had a successful fine-tuning of ResNet50, yielding a Validation accuracy of 89.8% and an AUC of 0.96 which pointed to the effectiveness of fine-tuning a pre-trained model for specific applications like deepfakes.Wu et al. [9] researched transfer learning as well using EfficientNet, obtaining a Validation accuracy of 89.1% and an AUC of 0.93. While it is advantageous to utilize pre-trained models to boost accuracy, it could be noted that, once again, no one has achieved DenseNet121 results. Other papers, like Yang et al. [10], explored fine-tuning strategies for deepfake detection and achieved a Validation accuracy of 90.2% and an AUC of 0.94. The DenseNet121 model, when fine-tuned for deepfake detection, achieved a validation accuracy of 97.32% and an AUC of 0.9982, outperforming existing models and demonstrating superior accuracy and robustness in identifying manipulated media. Table I summarizes key prior works on CNN-based deepfake detection, highlighting their methods, reported validation accuracies (Val Acc), AUC, and limitations, along with a comparison to our proposed DenseNet121 model.

TABLE I: Compact Summary of CNN-Based Deepfake Detection Studies

REF	Method	Val Acc	AUC	Limitations
[1]	CNN-based Model	91.5%	0.92	Limited performance in real-time detection; lacks robustness across different datasets
[2]	ResNet-50	88.3%	0.94	Requires high computational resources; limited performance on small datasets
[5]	Hybrid CNN (ResNet + MobileNetV2)	89.6%	0.95	Computationally expensive; accuracy drops with complex inputs
[3]	DenseNet121	97.32%	0.99	High computational demand for real-time processing; needs specialized hardware
[6]	EfficientNet-B7	88.5%	0.94	Lower performance compared to DenseNet; struggles with feature reuse
[7]	EfficientNet	87.2%	0.92	Less effective than DenseNet in feature reuse; less robust to small dataset sizes
[8]	ResNet50 Fine-tuned	89.8%	0.96	Limited feature extraction compared to DenseNet121; struggles with high variability in deepfakes
[9]	EfficientNet (Transfer Learning)	89.1%	0.93	Still lags behind DenseNet121 in terms of **AUC** and **Val Acc**
[10]	Fine-tuned CNN Model	90.2%	0.94	High resource requirements for fine-tuning; lower performance on complex deepfakes
Proposed Model	DenseNet121 (Fine-Tuned)	**97.32%**	**0.99**	High computational demand; optimal performance with large datasets and powerful hardware

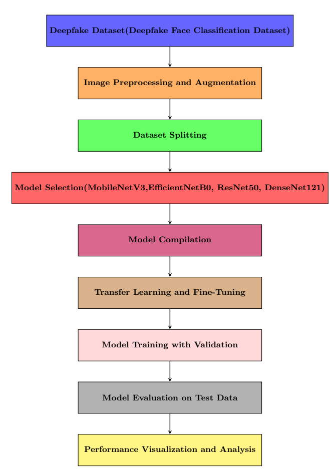

Fig. 1: A Workflow of Methodological Steps

III. Methodology

The methodology of this study, including the learning and evaluation of deep learning models for the purpose of deepfake image detection, is the primary focus. The results of four CNNs (i.e., DenseNet121, ResNet50, EfficientNetB0, and MobileNetV3) will be compared directly in terms of accuracy, AUC, computational complexity, and computational efficiency. Figure 1 provides a visual representation of the methodological workflow.

A. Dataset Description

The dataset used in this study is the Deepfake Face Classification Dataset available on Hugging Face [11]. The dataset consists of 31,120 images that are split into 2 labels, fake and real. The images are organized into three directories, the train directory, the val directory, and the test directory. The training directory includes 25,696 images and is organized with 12,848 fake images and 12,848 real images. The test directory includes 3,212 images, with 1,606 fake and 1,606 real images. In addition, the val directory also includes 3,212 images, with 1,606 fake images and 1,606 real images. The dataset consists of images with diverse representations in terms of lighting, facial expressions, and occlusions, which allows for better non-reliant models to be constructed. Figure 2 shows sample images that were collected from the dataset. The dataset is summarized in Table II.

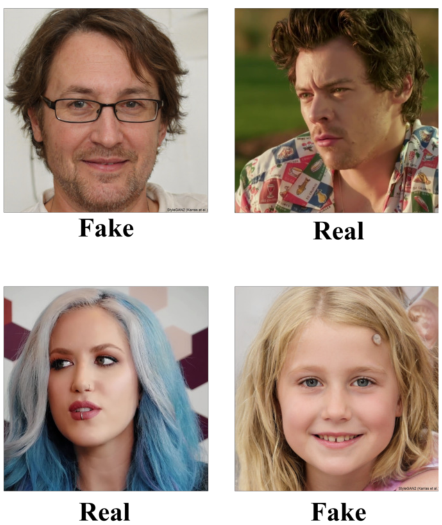

Fig. 2: Sample Images from Dataset

TABLE II: Dataset Distribution for Deepfake Detection

Dataset Split	Fake Samples	Real Samples
Training	12,848	12,848
Validation	1,606	1,606
Test	1,606	1,606
Total	25,696	25,696

B. Data Preprocessing

The image datasets are organized into two classes: Deepfake and Real, and the images are organized in folders for training, validation, and test datasets. In order for the model to receive the same input data, every image has been resized to 256px by 256 px. Each image is normalized by dividing every pixel value by 255, so every pixel value is between 0-1. Each time an image is passed to the model, it is in a batch of image data with a batch size of 32, both improving memory and efficiency when training.

For training, a set of augmentations are applied for the model to improve robustness. The basis for augmentations are geometric transformations, random horizontal flips, random rotations, and random perspective distortions, which will change the position and orientation of faces within the images. The Photometric Transformations are applied: random color jitters (brightness, contrast, saturation), random grayscale images, and Gaussian blur help to simulate the lighting changes and the distortions that can appear in the real world. Lastly, to replicate the various qualities of real images, there are Sharpness Adjustments, randomly sharpening or softening images.

TABLE III: Experimental Setup and Parameters for Deepfake Detection

Parameter/Setup	Details
Dataset	Deepfake Face Classification Dataset (Hugging Face)
Classes	Deepfake, Real
Image Size	256x256 pixels
Batch Size	32
Epochs	20
Optimizer	Adam (initial learning rate = 1e-3)
Loss Function	Binary Cross-Entropy
Evaluation Metrics	Accuracy, AUC, Precision, Recall, F1-score
Pre-trained Models	DenseNet121, ResNet50, EfficientNetB0, MobileNetV3
Model Modifications	Data Augmentation, GAP $\rightarrow$ Dense(512, ReLU) $\rightarrow$ Dropout(0.5) $\rightarrow$ Dense(256, ReLU) $\rightarrow$ Softmax (2-class)
Data Augmentation	Rotation (±20°), Shifts (10%), Shear (10%), Zoom (20%), Flip, Brightness (10%)
Normalization	Pixel values rescaled to [0,1]
Framework	PyTorch / torchvision
Programming Language	Python
Libraries	NumPy, OpenCV, Matplotlib, scikit-learn, timm
Hardware	GPU (NVIDIA CUDA-enabled)
Environment	Jupyter Notebook / Google Colab

C. CNNs Architecture Details

1) ResNet50: The ResNet50 model accepts residual connections, which can help address the vanishing gradient problem with deep networks. ResNet enables the network to learn residual mappings by using skip connections that bypass one or more layers. This is helpful when learning complex features in the data and is critical for a deep architecture such as deepfake detection.

The block operation in ResNet is represented as:

$$Y = X + F(X, W) \tag{1}$$

2) MobileNetV3: MobileNetV3 makes use of squeeze-and-excitation blocks and hard-swish activations for a better performance and efficiency combination, which are essential for deepfake detection tasks that require reliable performance without excessive computational resources.

The depthwise separable convolution operation for **MobileNetV3** can be written as:

$$Y = (X \odot W_{\text{depthwise}} + b_{\text{depthwise}}) \cdot W_{\text{pointwise}} + b_{\text{pointwise}} \tag{2}$$

3) EfficientNetB0: The EfficientNet architecture uses finite compound scaling, meaning it can scale the depth, width, and resolution of a network in a balanced way. EfficientNet uses a combination of depthwise separable convolutions and smart scaling to find a good balance between accuracy and efficiency, which is ideal for deepfake detection, especially when resources are limited.

The convolution operation in EfficientNet can be defined as:

$$Y = \text{Conv}(X, W) + b \tag{3}$$

4) DenseNet121: DenseNet121 connects each layer to every other layer; every layer is able to use all previously computed features. The dense connectivity allows for proper avoidance of degeneration and improves the flow of gradients as well as reducing overfitting. DenseNet121 uses all previous feature computations, allowing for efficient reuse of the computed features, and helps DenseNet121 to perform better when using smaller data sets. In DenseNet121, each layer utilizes the computed features from all preceding layers. Therefore, DenseNet121 improves representations and ultimately performance.

The operation for DenseNet121 can be represented as:

$$Y_l = \text{Concat}(X_1, X_2, \ldots, X_{l-1}) \cdot W_l + b_l \tag{4}$$

The DenseNet121 architecture uses dense connections to give layers the ability to connect using all outputs of all previous layers as inputs, which allows the most effective use of features. A convolutional layer is first defined, as well as a Max Pooling layer, to reduce size, and then four dense blocks - six layers of convolutions with ReLU activation, twelve layers of convolutions, twenty-four layers of convolutions, and sixteen layers of convolutions. After each dense block is a Transition Layer, which reduces the feature map and channel counts, the transition layer does this with 1x1 convolutions and 2x2 average pooling. The final component of the DenseNet121 model is a Global Average Pooling (GAP) layer, which transforms the feature maps into a single vector for use in fully connected layers. The model then makes its classification decision using a softmax output to classify the images as either Deepfake or Real.

Figure 3fshows the proposed model's layer architecture, while Table IV shows layer details.

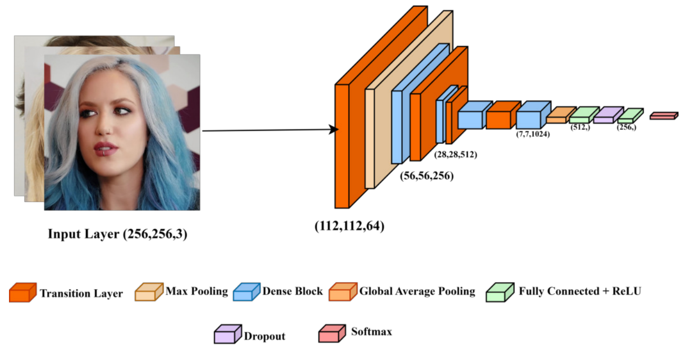

Fig. 3: DenseNet121 Model's Layer Architecture

TABLE IV: Layer Details of the DenseNet121 Architecture

Layer Type	Configuration	Output Shape
Input Layer	256 x 256 x 3 (RGB image)	(256, 256, 3)
Conv1 (Initial Convolution)	64 filters, 7x7 kernel, stride 2, ReLU activation	(112, 112, 64)
Max Pooling	3x3 pool size, stride 2	(56, 56, 64)
Dense Block 1	6 layers, each with 32 filters, 3x3 kernels, ReLU activations	(56, 56, 256)
Transition Layer 1	1x1 convolution, 2x2 average pooling, 128 filters	(56, 56, 128)
Dense Block 2	12 layers, each with 32 filters, 3x3 kernels, ReLU activations	(28, 28, 512)
Transition Layer 2	1x1 convolution, 2x2 average pooling, 256 filters	(28, 28, 256)
Dense Block 3	24 layers, each with 32 filters, 3x3 kernels, ReLU activations	(14, 14, 1024)
Transition Layer 3	1x1 convolution, 2x2 average pooling, 512 filters	(14, 14, 512)
Dense Block 4	16 layers, each with 32 filters, 3x3 kernels, ReLU activations	(7, 7, 1024)
Global Average Pooling	Global average pooling layer	(1024,)
Fully Connected	512 units, ReLU activation	(512,)
Dropout	Dropout rate: 0.5	(512,)
Fully Connected	256 units, ReLU activation	(256,)
Output Layer	Softmax (2 classes: Deepfake, Real)	(2,)

IV. RESULTS AND PERFORMANCE ANALYSIS

On the Deepfake Face Classification Dataset, the evaluated architectures were trained and validated, with images classified as either "Deepfake" or "Real." The accuracy and AUC metrics were monitored during training and validation to assess the model's performance and identify any overfitting. The weights of the models that performed the best based on the highest validation accuracy and AUC were preserved after the training was completed.

The model based on DenseNet121 (proposed) achieved a training accuracy of 97.65% and a validation accuracy of 97.32%. The training loss was 1.58% and the validation loss was 2.01%. The AUC for the proposed model was 99.82%, showing a stronger classification performance. These results show that the model learned how to successfully separate deepfake and real images, with only a slight incidence of overfitting and a satisfactory generalization ability for new validation data.

ResNet50 achieved a validation accuracy of 92.15% and an AUC of 98.82%. Despite its adequate performance, DenseNet121 outperformed this model in both accuracy and AUC. EfficientNetB0 and MobileNetV3 had similar results, with validation accuracies of 97.32% and AUCs of 99.44% and 99.77%, respectively, but they were not as consistent or effective as DenseNet121.

All models performed well, but the DenseNet121 architecture outperformed the others in terms of both validation accuracy and AUC, making it the most reliable model for deepfake detection in this study. Figures 4 and 5 present the metrics for high and low accuracy, as well as the loss percentage, for a

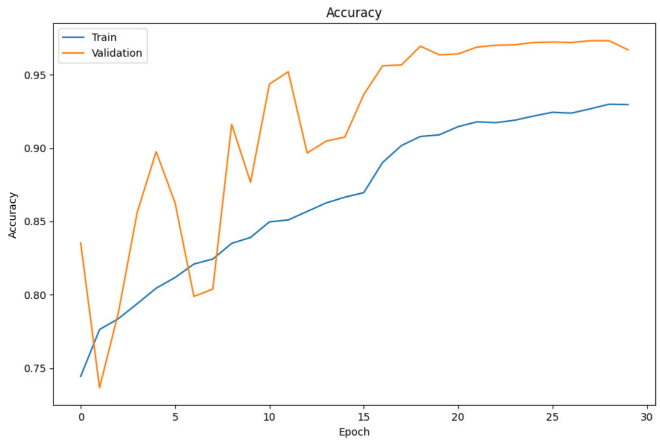

Fig. 4: Accuracy Curve Of Fine-Tuned DenseNet121 model

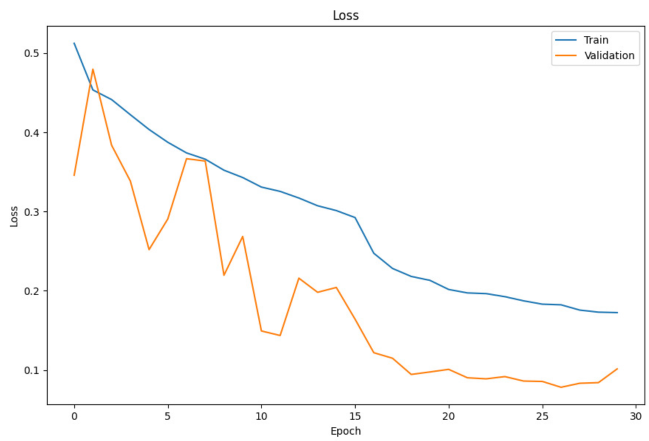

Fig. 5: Loss Curve Of Fine-Tuned DenseNet121 model

more visual comparison of model performance.

Figure 6 shows the confusion matrix for the fine-tuned Densenet121 model. The confusion matrix for the validation set indicates the model accuracy of classifying between Class 0 (Real) and Class 1 (Deepfake). The model successfully identified 1513 real images and 1593 deepfake images, relating to True Negatives (TN) and True Positives (TP), respectively. In this process, some images were wrongly classified: 93 real images scored as deepfakes (False Positives, FP), and 13 deepfake images as real (False Negatives, FN). Overall, the model performs very well, with a high number of correct predictions and minimal errors, which indicates that it can generalize to unseen validation data.

The ROC curve for the DenseNet121 model demonstrates its excellent performance in distinguishing between deepfake and real images. The curve shows a sharp increase in the True Positive Rate (TPR) with a low False Positive Rate (FPR), indicating high classification accuracy. The model achieves an AUC of 0.9982, which reflects its strong ability to classify the images correctly. This suggests that DenseNet121 is highly effective in deepfake detection, with minimal misclassification. Figure 7.

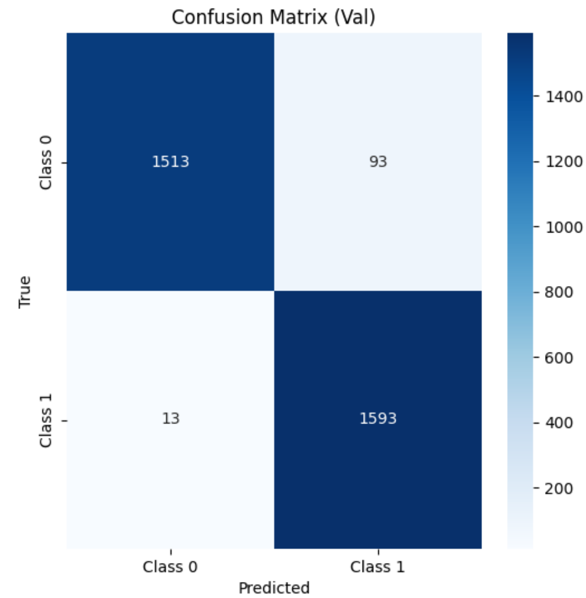

Fig. 6: Confusion Matrix of Fine-Tuned DenseNet121 Model

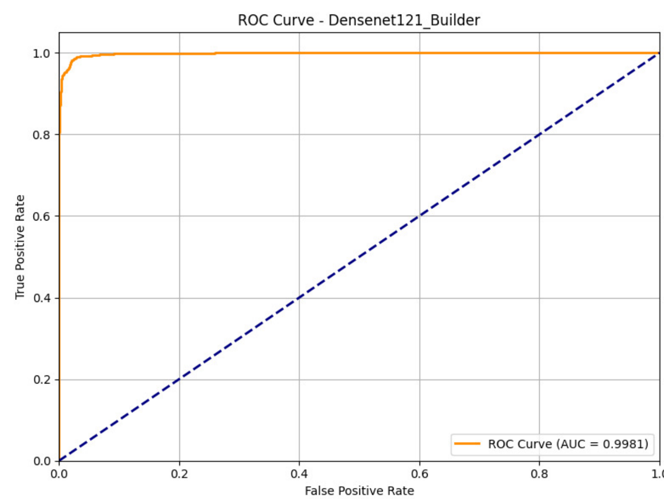

Fig. 7: ROC Curve of Fine-Tuned DenseNet121

The DenseNet121 Model outperformed all other architectures tested in its classification of deepfake versus real images, obtaining a validation accuracy of 97.32% and an AUC of 99.82%. Not only did this model have the greatest accuracy, but it also obtained the greatest AUC, when the other models were compared. EfficientNetB0 (validation accuracy = 97.32% with a 99.44 AUC) and MobileNetV3 (validation accuracy = 97.32% with 99.77 AUC) came in second place, and ResNet50 came in third with only 92.15% accuracy and an AUC of 98.82%. All results are displayed in Table V, and the AUC of each model is also shown in Figure 8.

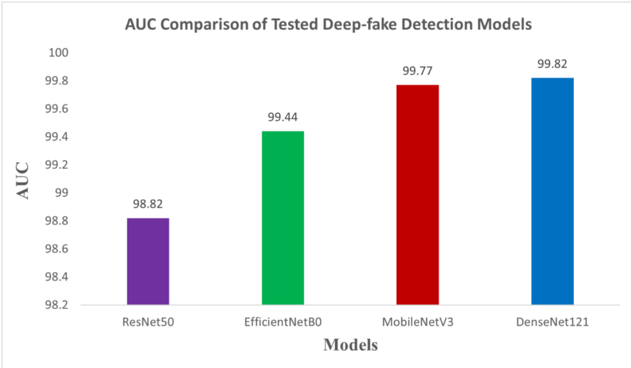

Fig. 8: Graphical Comparison of all Models

TABLE V: Accuracy and AUC Comparison of Tested Deep-fake Detection Models

Model	Validation Accuracy	AUC
ResNet50	92.15%	98.82%
EfficientNetB0	97.32%	99.44%
MobileNetV3	97.32%	99.77%
Proposed Model (DenseNet121)	**97.32%**	**99.82%**

Recent developments in deepfake detection models have led to three outstanding real-time learned models: DenseNet121, which has been fine-tuned specifically for deepfake detection and demonstrates a validation accuracy of 97.32% and an AUC of 0.9982, making it arguably one of the best models to rely on for this purpose [3]. Secondly, ResNet50, which achieved a validation accuracy of 88.3% and an AUC of 0.94 [2]. Then EfficientNet, achieving 87.2% validation accuracy and an AUC of 0.92 [7]. Hybrid CNN architectures, e.g., ResNet + MobileNetV2, have also shown promising results with a validation accuracy of 89.6% and an AUC of 0.95 [5]. Overall, hybrid CNN architectures were still not as good as fine-tuned DenseNet121, as it can be clearly seen from TableVI that the fine-tuned DenseNet121 model has significantly outperformed all other architectures with regard to accuracy and AUC score. Thus, both common competitor models have demonstrated that fine-tuned DenseNet121 is the best-performing model for deepfake classification, especially when reducing computational burden on large datasets combined with time and processing.

TABLE VI: Comparative Analysis of Deepfake Detection Models Based on Recent Studies

Model (Ref.)	Validation Accuracy	AUC
[1] CNN-based Model	91.5%	0.92
[2] ResNet-50	88.3%	0.94
[5] Hybrid CNN	89.6%	0.95
[3] DenseNet121	97.32%	0.99
[6] EfficientNet-B7	88.5%	0.94
[7] EfficientNet	87.2%	0.92
[8] ResNet50 Fine-tuned	89.8%	0.96
[9] EfficientNet (Transfer Learning)	89.1%	0.93
[10] Fine-tuned CNN Model	90.2%	0.94
Proposed Model: (Fine-Tuned DenseNet121)	**97.32%**	**0.99**

V. Discussion and Conclusion

Along with the improved DenseNet121 model, it continuously checks how well other strong CNNs (ResNet50, EfficientNetB0, MobileNetV3) perform in detecting deepfakes. The optimal fine-tuning of DenseNet121 achieved the highest performance, with a validation accuracy of 97.32% and an AUC of 0.9982, indicating better performance and more robustness than other models. The dense connectivity of DenseNet121, which allows features to be reused across the layer, was particularly important for a more efficient one-and-done pipeline to detect deep fakes; this network achieved better performance compared to the ResNet50 (with a validation accuracy of 92.15% and an AUC of 0.9882) and EfficientNetB0 (with a validation accuracy of 97.32% and an AUC of 0.9944) models. These findings also highlight the advantage of fine-tuning pre-trained models like DenseNet121 for the specific task of deepfake detection. The performance of the model in discriminating between real and deepfake was very satisfactory with minimal misclassifications and the model showed excellent generalization to unseen data. Nevertheless, the computational requirements of DenseNet121 for real-time processing remain high, indicating that future optimization is necessary. This work is one of the first efforts for the identification of personal Vlogs and represents a significant contribution to media forensics and possible uses can be found in social media, cybersecurity and digital content verification. In the future, we can enhance the computational efficiency of the model and expand the dataset for cross-validation.

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